

HACKDAY 1.0 · TECH FOR A BETTER TOMORROW

RakshaCall



Real-Time Scam-Call Protection Assistant

Detect. Explain. Warn. Protect.

Protecting citizens from high-pressure social engineering, digital arrest, and financial extortion while the call is happening — not after funds are lost.

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 **Live Demo:** daanialmirza5.github.io/RakshaCall/

**RakshaCall**
Scam Call Protection

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REAL-TIME SCAM CALL PROTECTION

Stay ahead of scam calls.

RakshaCall detects high-risk social-engineering patterns during suspicious calls and helps you understand what's happening before you take action.

Start Protection

Try Live Demo

Privacy-first · Explainable alerts · Built for Indian scam patterns

**RakshaCall**

Protection Active

Current Risk Level

High Risk

This strongly matches a known scam pattern.

Detected Tactics



Claims to be a government/police authority

Real officials don't investigate over a phone call.



Threatening arrest or legal action

Genuine legal processes are never resolved by a phone call.



Telling you to stay silent or not hang up

A classic tactic to stop you from checking with family.



Asking you to transfer money

This is the actual goal of almost every such scam.

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Scam Calls Manipulate People in Real Time

Scammers don't just steal data — they psychologically coerce victims before they realize what is happening.

THE ANATOMY OF A PSYCHOLOGICAL SCAM CALL

SCAMMER → Authority Impersonation → Urgency & Threats → Isolation → OTP / Money / Remote Access → VICTIM

Predatory Modalities

- Digital Arrest: Fake CBI/Police video surveillance
- Fake Bank KYC: Urgent account block threats
- Customs / Courier: Seized illegal parcels
- SIM Deactivation: Immediate telecom shutoff
- Task / Investment: Upfront deposit extortion

Psychological Traps

- Enforced Secrecy: 'Do not tell family or police'
- High-Stress Panic: 'Transfer in 10 mins or arrest'
- Credential Extraction: OTPs, PINs, Passwords
- Targeted Victims: Seniors & non-technical citizens
- ~₹3,000+ Cr estimated lost to Digital Arrest in 2025 alone

Why Defenses Fail

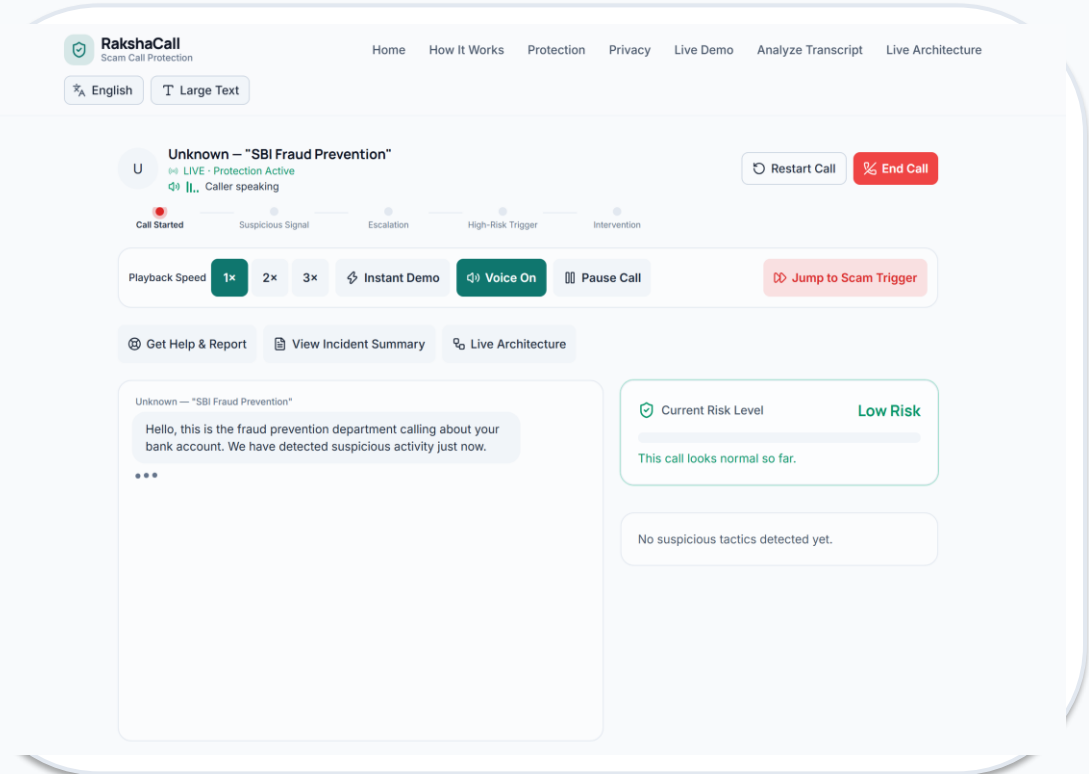
- Bank Fraud Alerts: Trigger AFTER money is moved
- Telecom Filters: Block spam numbers, not live voices
- Cybercrime Portals: Report incidents days later
- Zero Real-Time Guidance: Victim is alone during call
-  **Need protection DURING the manipulation!**

Meet RakshaCall

A conversational safety assistant that detects scam tactics and protects citizens while the conversation is happening.

CONVERSATION → DETECT → ASSESS RISK → WARN → PROTECT → HELP

- 🛡️ Real-Time Detection: Continuous stream evaluation of dialogue as words are spoken
- 🔍 Explainable Tactics: Clear classification across 8 social-engineering vector categories
- 🚨 Risk Escalation: Dynamic progressive threat scoring (0–100) across Low, Med, High
- 🛑 Safety Intervention: Full-screen actionable takeover: STOP · DO NOT SHARE · DISCONNECT
- 👨👩 Trusted Contact Check-In: One-tap emergency check-in alert to family or guardians
- 🇮🇳 Indian Scam Aware: Native recognition of English, Hinglish, Hindi, and Marathi



From Conversation to Protection

Zero black-box AI, zero cloud latency, and 100% on-device privacy.

1. Input

Call Stream
Mic Audio
Pasted Text

2. Normalize

Unicode
Transliteration
Case & Spacing

3. Signals

8 Multi-Vector
Scam Tactic
Categories

4. Scoring

Co-occurrence
Boost & Math
Weights (0–100)

5. Classifier

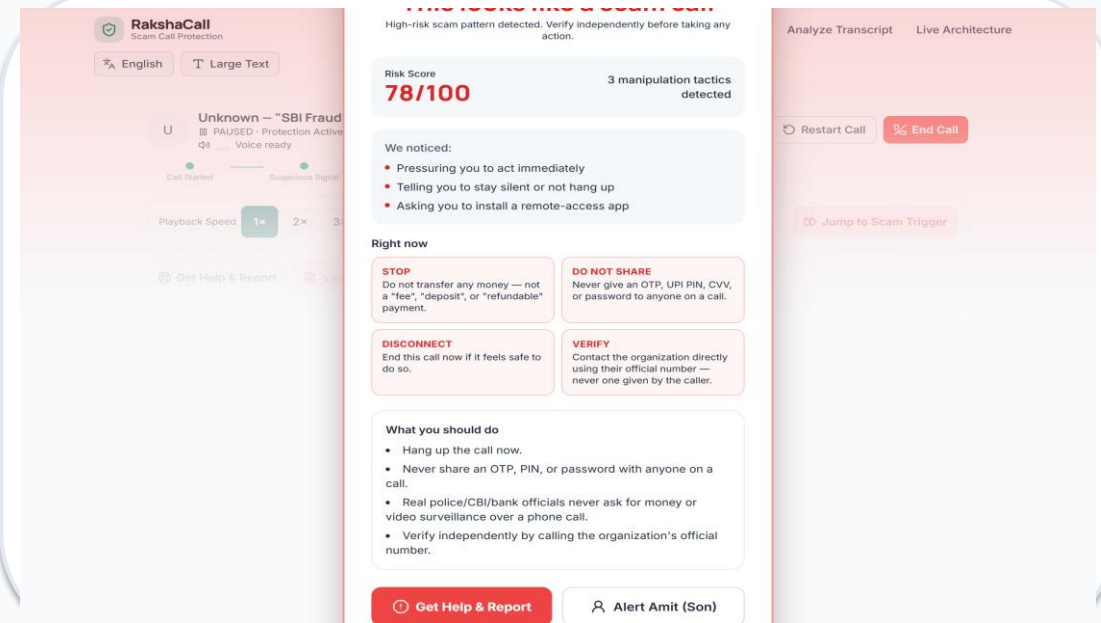
LOW (0–24)
MEDIUM (25–54)
HIGH (55–100)

6. Action

Screen Takeover
Trusted Alert
1930 Helpline

8 Observable Social-Engineering Categories

- Authority Impersonation: CBI, Police, Customs, RBI officials
- Urgency & Threat: Immediate arrest, bank account blocking
- Isolation & Secrecy: 'Do not hang up, do not tell family'
- Financial Demand: 'Refundable verification fees', penalties
- OTP & Credential Harvesting: UPI PIN, CVV, OTP requests
- Remote Access: AnyDesk, TeamViewer, QuickSupport requests



When Risk Becomes High, RakshaCall Acts

Breaking the psychological manipulation loop with four unambiguous citizen commands.



STOP

Do not transfer any money or pay verification fees.



DO NOT SHARE

Never disclose OTPs, PINs, CVVs, or passwords.



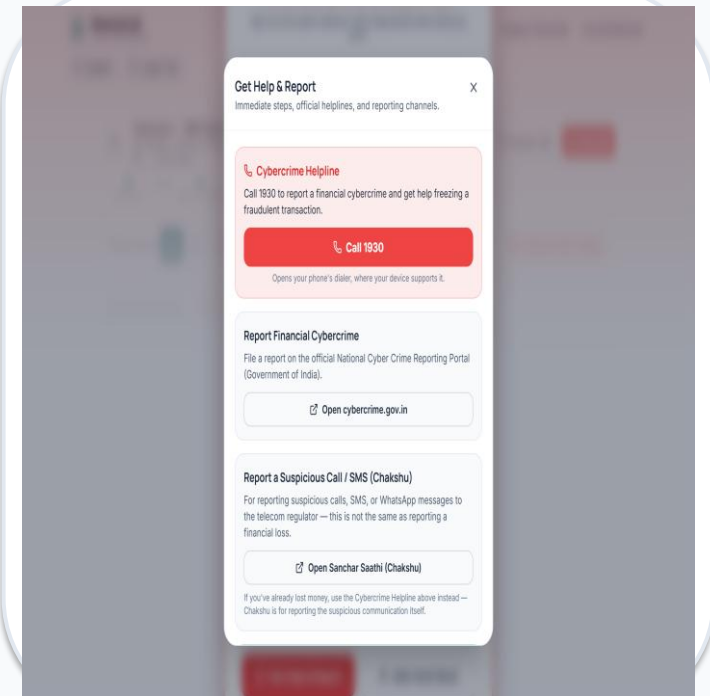
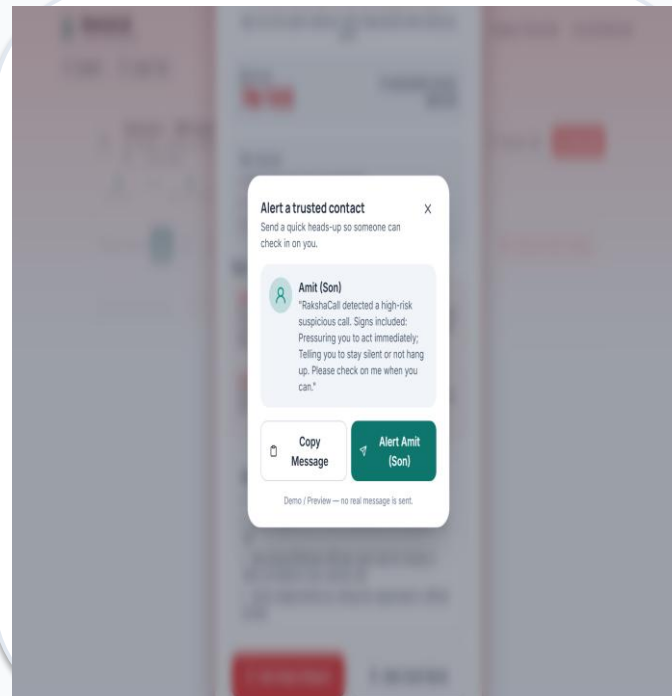
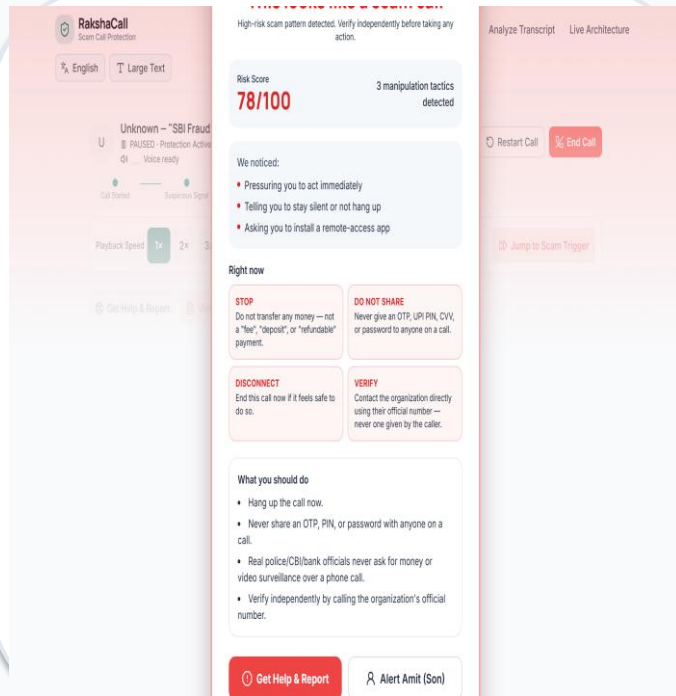
DISCONNECT

Immediately hang up the suspicious call.



VERIFY

Independently contact official numbers directly.



See What RakshaCall Is Doing — Live

A dedicated technical observatory providing live visibility into internal function execution without cluttering citizen UX.

Browser-Native Cross-Tab Synchronization

- Real Execution Graph: Reflects actual function calls, state transitions, and timestamps — not an animation.
- BroadcastChannel Pub/Sub: Runs in a separate tab or side-by-side window, synchronizing live with the active call session.
- 7 Observable Workflows:
 1. Live Call Protection (stream orchestration)
 2. Detection Intelligence (regex/math matrix)
 3. Voice & Speech Activity (TTS synthesis)
 4. Incident & Response (forensics builder)
 5. Manual Analysis (pasted text engine)
 6. Emergency Response (action triggers)
 7. Session Lifecycle (state reset & tear-down)
- Node-Level Inspections: Status (running/success), execution duration, inputs, and extracted quotes.


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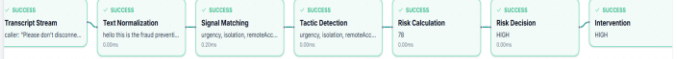
Live Architecture

A real-time observability view over RakshaCall's own execution — every node reflects an actual function call, not a scripted animation. This page shows the last session's real events; for a graph that updates live while you watch, open **Live Architecture** from within an active call or Paste Analyze, where it stays open alongside the real run.

Live Runtime Workflow
LIVE

Execution: RK-E225 · Client-side runtime · No backend telemetry

Live Call Protection
Detection Intelligence
Voice & Speech Pipeline
Incident & Response
Manual Analysis
Emergency & Protection Response



```

graph LR
    A[Transcript Stream] --> B[Text Normalization]
    B --> C[Signal Matching]
    C --> D[Tactic Detection]
    D --> E[Risk Calculation]
    E --> F[Risk Decision]
    F --> G[Intervention]
            
```

LIVE EXECUTION LOG

- 17:26:34 ✓ Protection Actions
- 17:26:24 ✓ Protection Actions
- 17:26:24 ✓ Intervention
- 17:26:24 ✓ Risk Decision
- 17:26:24 → Risk Decision
- 17:26:24 ✓ Risk Calculation
- 17:26:24 → Risk Calculation
- 17:26:24 ✓ Tactic Detection
- 17:26:24 → Tactic Detection
- 17:26:24 ✓ Signal Matching
- 17:26:24 → Signal Matching
- 17:26:24 ✓ Text Normalization
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- 17:26:24 ✓ Transcript Stream
- 17:26:23 ○ Intervention
- 17:26:23 ✓ Risk Decision
- 17:26:23 → Risk Decision
- 17:26:23 ✓ Risk Calculation
- 17:26:23 → Risk Calculation
- 17:26:23 ✓ Tactic Detection

RakshaCall — From Detection to Action

Protecting citizens during the scam call — when intervention matters most.

🌟 Core Hackathon Value

- ✓ Solves the 'Golden Hour' Dilemma: Intervenes before money leaves the victim's account.
- ✓ 100% Privacy-Preserving: Zero audio/text leaves the user's browser. Zero backend.
- ✓ Zero Cost & Zero API Dependencies: Immune to cloud outages, subscription costs, or API keys.
- ✓ Explainable & Transparent: Every risk score points to exact phrases and tactic categories.
- ✓ Multilingual Indian Context: Supports English, Hinglish, Hindi, and Marathi code-mixing.
- ✓ 203/203 Vitest Suite Passed: Production-tested reliability and complete CI/CD automation.

🚀 Verified Live Resources

- **Live Application Demo:**
<https://daanialmirza5.github.io/RakshaCall/>
- **Live Runtime Observatory:**
<https://daanialmirza5.github.io/RakshaCall/#/observatory>
- Open Source Repository:
<https://github.com/daanialmirza5/RakshaCall>
- Verified Demo Video:
<docs/demo/RakshaCall-Demo.mp4>

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